

Centre for Credible AI · Warsaw

AI as a Mathematician

Open Erdős problems, research-level prompting,
and the next mathematical workflow

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Warsaw · Friday, 22 May 2026

research-grade reasoning data + evals

UCL AM

The field report

A personal sprint through open Erdős problems with frontier reasoning models

Research math is changing from scarce proof generation to abundant proof candidates.

13

solution claims
(updated list)

4

almost / partial
results

3

previously
verified

500+

problems triaged
in the marathon

Important stance: these are claims until proof digestion, literature review, and community/formal verification catch up.

Why Erdős problems?

A public, diverse, community-curated testbed for research reasoning

public problem market

Erdős Problems now act like a shared arena where progress, partial progress, and failure modes are visible.

“not a finely calibrated benchmark”

— Erdős AI contributions wiki caveat

The benchmark qualities

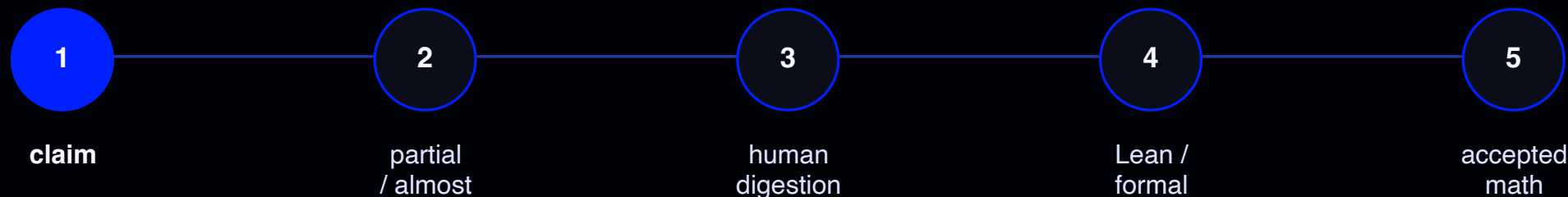
- short statements with deep surrounding literature
- diverse domains: number theory, graphs, analysis, geometry, set theory
- public pages track status, comments, references, and sometimes Lean formalization
- failures are informative: missing hypotheses, prior art, or wrong formulations

The current claim portfolio

Keep the taxonomy visible; it protects credibility

13 solution claims

#522 #603 #610 #750
#856 #888 #896
#953 #956 #1092
#1133 #1151 #1190



Partial / almost: #514 · #741 · #906 · #1201

Previously verified: #258 · #858 · #1148

Best phrasing on stage: “AI-assisted solution claims under verification,” not “AI solved everything.”

PROVED (LEAN)

Let $a_1, a_2, \dots$ be a sequence of positive integers with $a_n \rightarrow \infty$. Is

$$\sum_n \frac{\tau(n)}{a_1 \cdots a_n}$$

irrational, where $\tau(n)$ is the number of divisors of n ?

#258: [ErGr80,p.62][Er88c,p.103]

irrationality

Erdős and Straus [ErSt71] proved this is true if a_n is monotone, i.e. $a_{n-1} \leq a_n$ for all n . Erdős [Er48] proved that $\sum_n \frac{\tau(n)}{t^n}$ is irrational for any integer $t \geq 2$.

Chojacki and GPT-5.4 Pro have answered this question in the affirmative, using the work of Tao and Teräväinen [TaTe25] (see [69] and [248]).

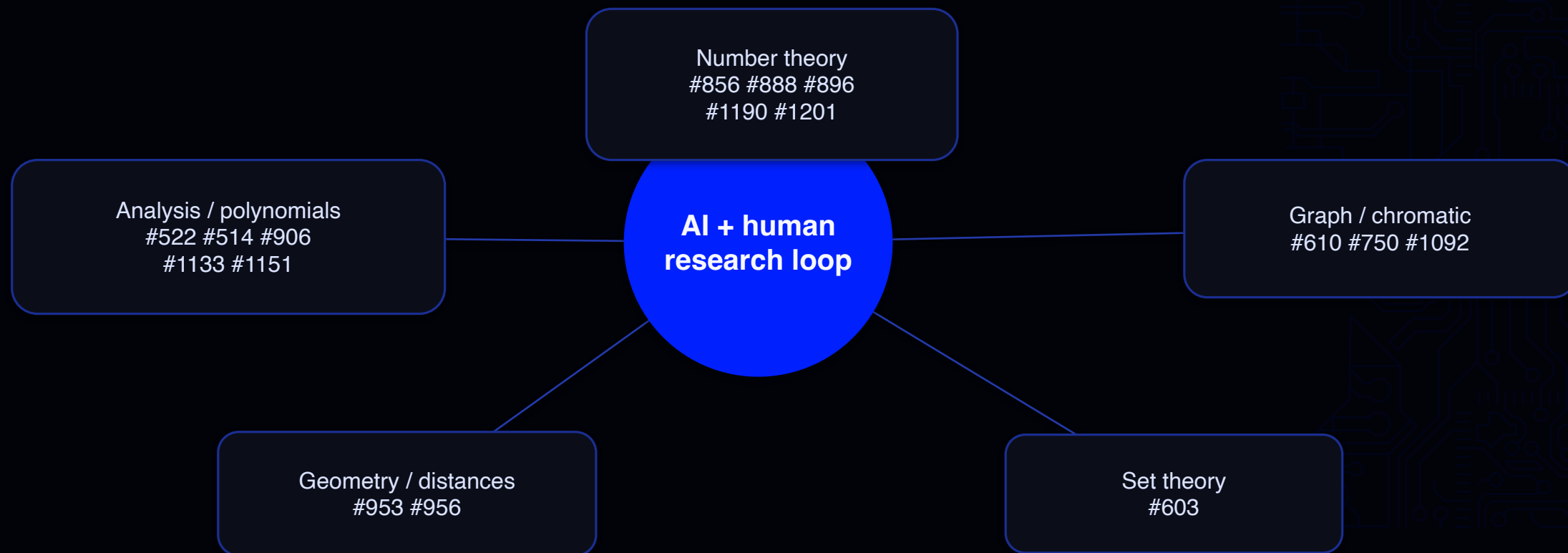
Very nice! This by the way is essentially the same sort of irrationality strategy that was used to resolve [1051], namely to show that, infinitely often, the tail decays faster than the reciprocal of the lcm of all previous denominators.

TerenceTao — 17:33 on 14 Apr 2026 | Reply 🍌3 🍌0 🍌0

The mathematical footprint

The sprint was not one trick; it crossed domains

The interesting signal is breadth: a single workflow touched several Erdős “neighborhoods.”

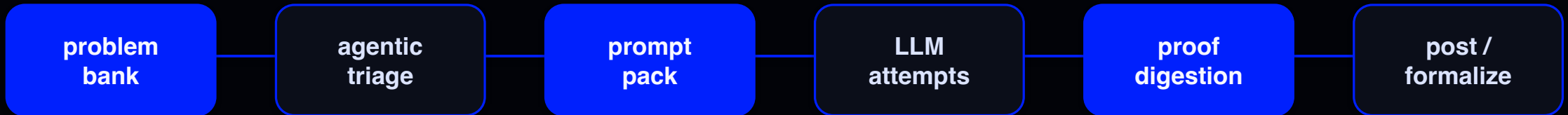


Model strength: synthesis across literature, examples, and proof styles.

Model weakness: it still needs a verifier and a mathematician to decide what matters.

The marathon workflow

The proof candidate is a late-stage artifact, not the first ask



What “agentic” did before the proof:

- pull exact statement and variants
- collect references, known bounds, nearby problems
- run small computations / sanity checks
- turn a vague problem into a structured context pack

The LLM becomes strongest after the problem has already been cleaned.

A raw statement is not a prompt. A prompt is a compressed research environment.

Prompting LLMs for research mathematics

Do not ask for “the answer”; ask for a disciplined research process

A good research prompt forces the model to expose its proof architecture before it writes prose.

1

State the exact gap

Known results, variants, hypotheses, and what remains open.

2

Propose proof architectures

At most three attacks; each with checkpoints that can fail.

3

Write lemmas, not vibes

Every step has a local claim, dependencies, and a failure mode.

4

Run an adversary

Ask the model to break its own proof before you read it.

Prompt skeleton

“Before solving, give the weakest known result and the precise gap.

Then propose up to three proof architectures. For each, name the theorem you need, the obstruction, and one falsifiable checkpoint.

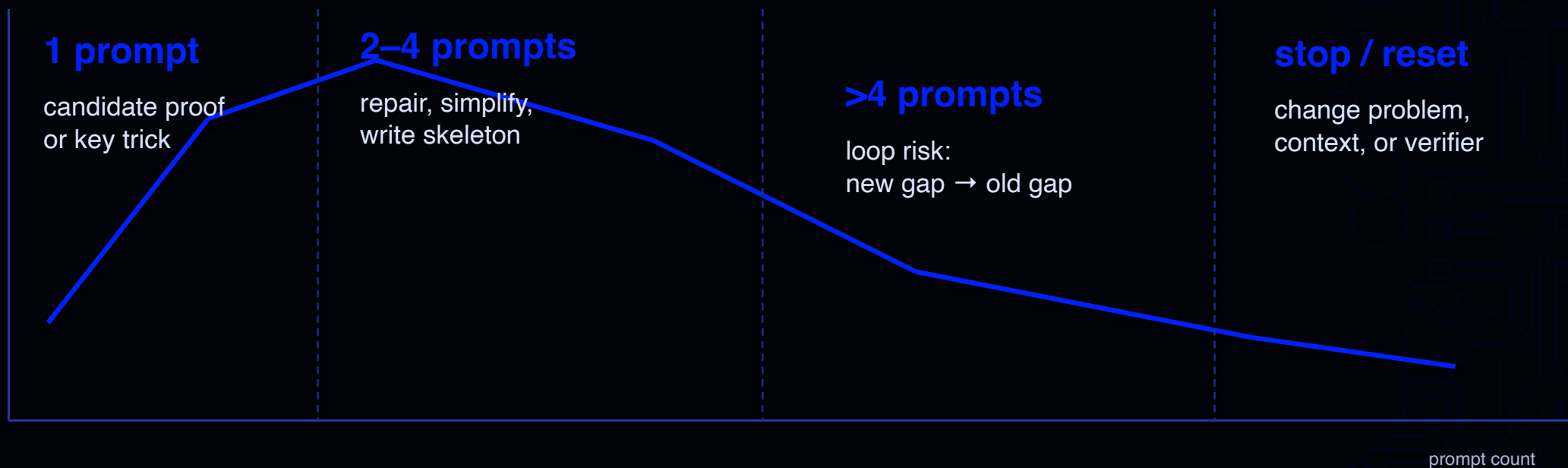
Only after that, write the proof skeleton and attack it adversarially.”

The “four prompt” heuristic

Long chats often start repairing their own repairs

What I found useful: optimize for fast, coherent hits — not heroic prompting.

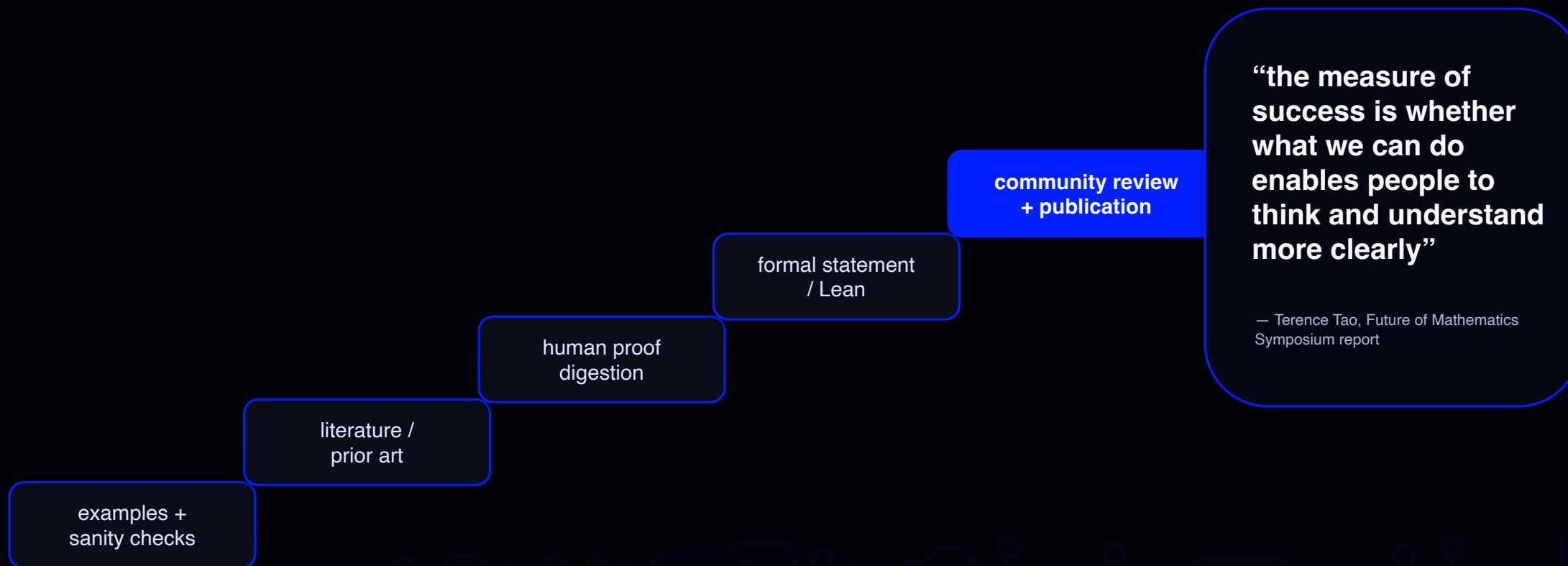
stability



Interpretation: if the model cannot produce a stable proof architecture quickly, the missing ingredient is usually outside the chat.

Verification ladder

A “proof-shaped object” is not yet a theorem



Credibility is a workflow property.

The harder the problem, the less we should trust a single opaque transcript.

Data from doing math

Verified research trajectories can train proof critics, judges, and RLVR systems

The next frontier is not “more final answers.” It is proof-process data.



Signal captured

what was justified · what was conditional · what was wrong · where the proof stopped

A verified trajectory is not just an answer; it is a supervised record of how research reasoning succeeded, failed, and recovered.

From proof generation to proof digestion

The bottleneck moved — and the social norms have to move with it

“Can it produce a proof?” is becoming the wrong first question.

proof-shaped
object

understood
theorem

Tao’s “pleasant dinner” analogy on the Erdős forum: in an age of proof abundance, quality and digestion matter more than dumping food on the table.

New review questions

- Is the statement exactly right?
- Where does the proof rely on known theorems?
- Which lemmas are genuinely new?
- Can it survive formalization or public scrutiny?

The external signal: Gowers

Research-level AI is visible outside the Erdős ecosystem

“

**PhD-level research in an hour or so,
with no serious mathematical input
from me**

— Timothy Gowers on ChatGPT 5.5 Pro

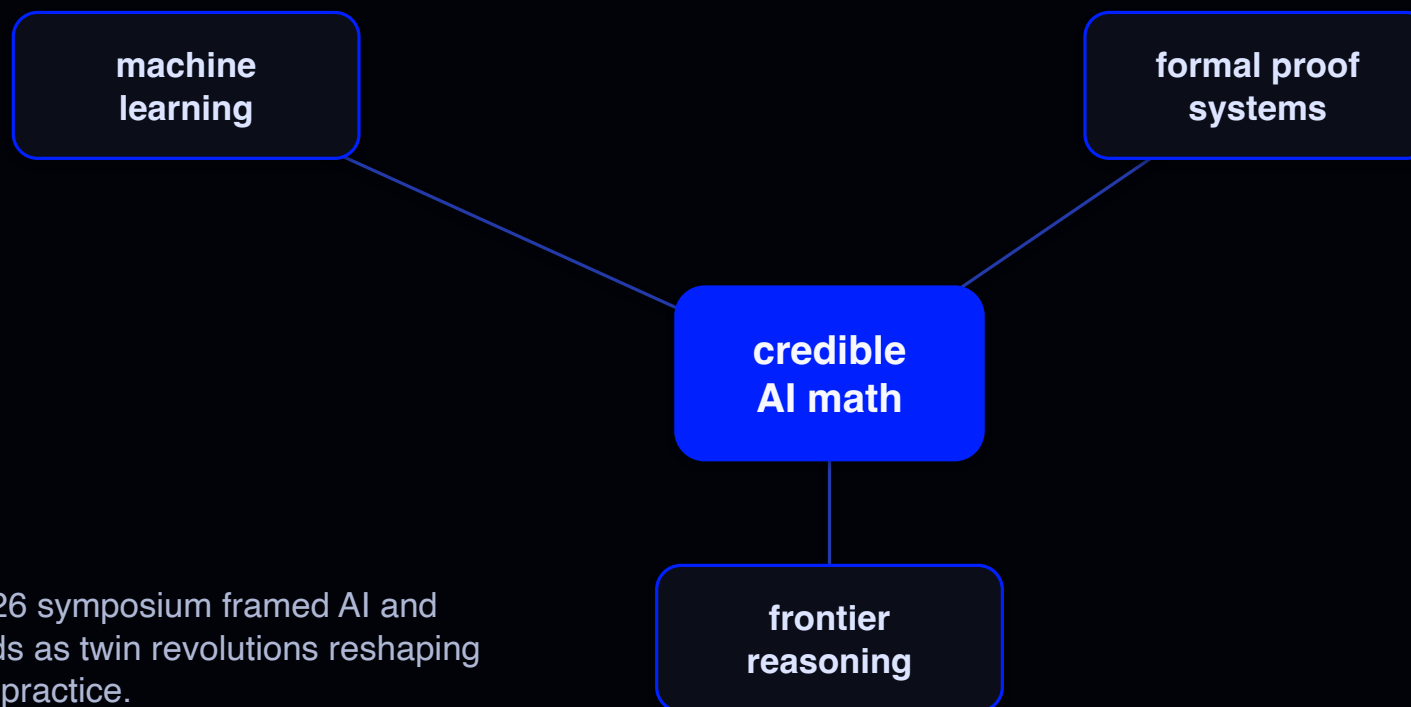
Why this matters

- publishable-looking contribution, not a contest answer
- human role: prompt control, assessment, communication
- open question: where do AI-produced results live, and how are they certified?

The field is organizing around the same question

Future of Mathematics: machine learning + formal methods + frontier reasoning

Mathematics + AI + formal proof workflows



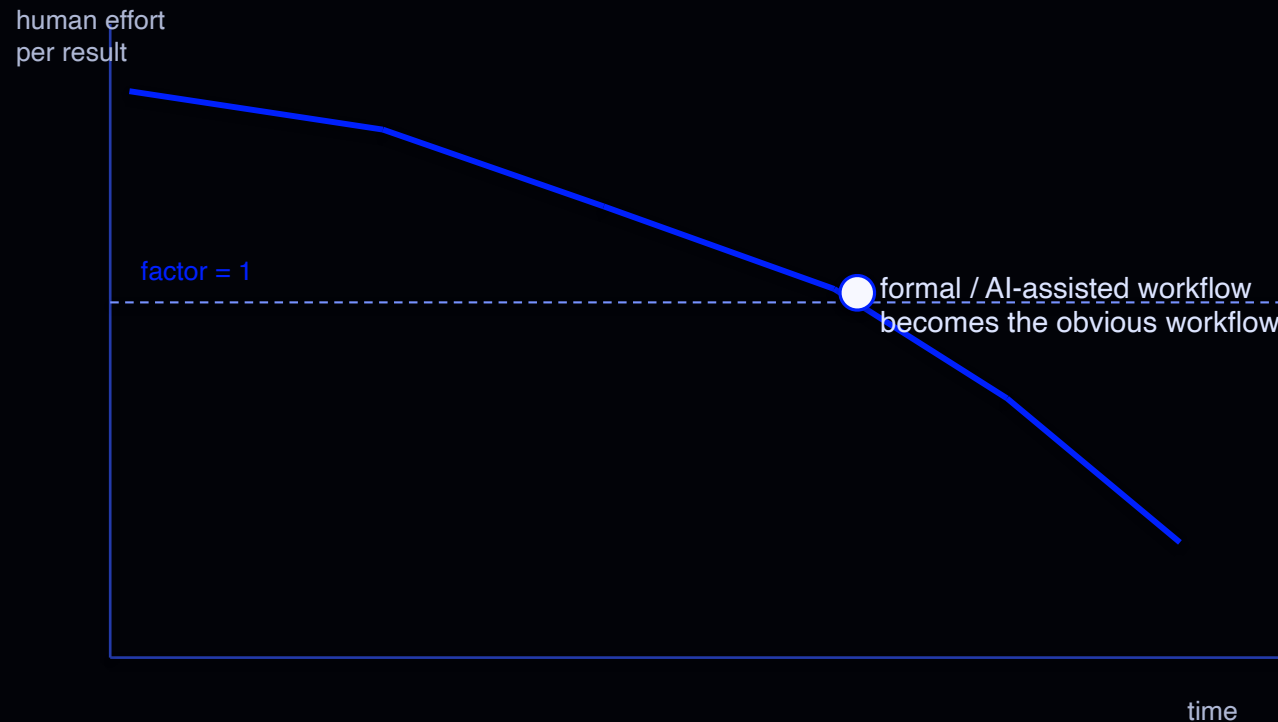
Stanford's 2026 symposium framed AI and formal methods as twin revolutions reshaping mathematical practice.

Centre point for credible AI: make outputs understandable, verifiable, and controllable — especially at the boundary of human review capacity.

What to expect next

The adoption curve starts when the workflow becomes faster, not merely possible

Kontorovich's useful lens: the "formalization factor" drops below 1 when formal work is faster than informal work.



Near-term shifts

- bulk triage of public problem banks
- automated literature/context packs
- proof critics trained on trajectories
- formalization as the default credibility filter

What remains human?

The “move 37” of mathematics is not just a proof

A frontier model can find arguments. A mathematician decides what becomes a theory.

My move 37 in mathematics would actually be a new definition, not a proof.

?

**choose questions
that matter**

∂

**invent concepts
and definitions**

□

**decide when proof
becomes
understanding**

∞

**build norms
for credit + review**

AI accelerates proof search so humans can spend more time on taste, abstraction, and explanation.

Credible AI mathematics

Three questions for this room

- 1 What should count as a credible AI-generated mathematical claim?
- 2 What proof-process data will train the next generation of reasoners?
- 3 How do we make AI help mathematicians understand more, not less?

High-throughput proof generation is here. The credible workflow is

provenance → digestion → verification → data

Appendix: exact current list

13 solution claims

#522, #603, #610, #856, #888, #896, #953, #956, #1092, #1133, #1151, #1190, #750

4 almost / partial

#514 — finished with another person

#741 — cleaning what is already proven from OpenAI / DeepMind

#906 — completed by someone else first; a bit different

#1201 — partial, but close

3 previously verified

#258

|

#858

|

#1148